

dot products

- only w/ linear transformations can we understand the dot product

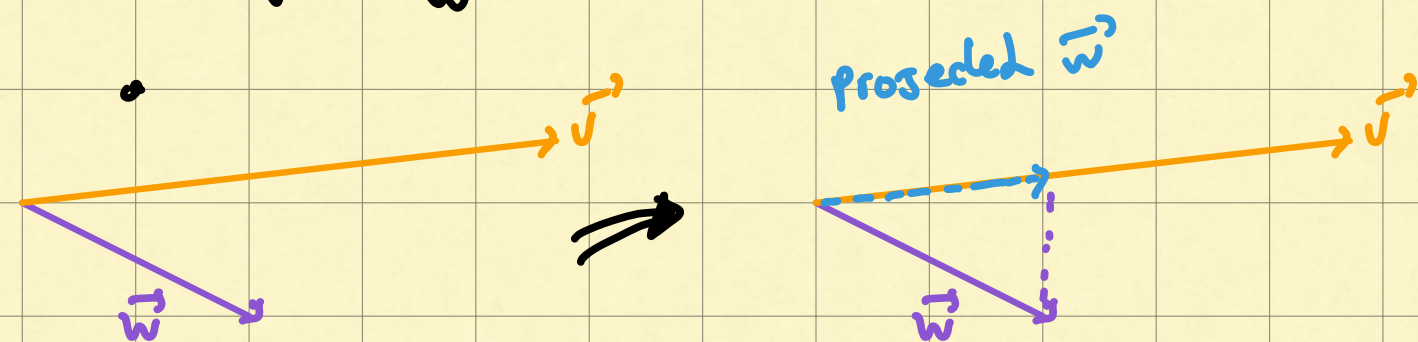
- standard view :

- given 2 vectors of same dimension

- dot product = putting up each coordinate, multiply them, adding result

ex: $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} d \\ e \\ f \end{bmatrix} = ad + be + cf$

$\vec{v}$ $\vec{w}$



multiply length projection onto : $\vec{v} \cdot \vec{w} = (|\text{proj}_{\vec{v}} \vec{w}|) \cdot (|\vec{v}|)$

length of $\vec{v}$, = $\vec{v} \cdot \vec{w}$

- dot product negative if vectors pointing opposite
- perpendicular, dot product zero, bc proj of 1 onto the other is zero vector

- this proj stuff treats the two vectors very differently

- order does not matter:

Proj $\vec{w}$ onto $\vec{v}$

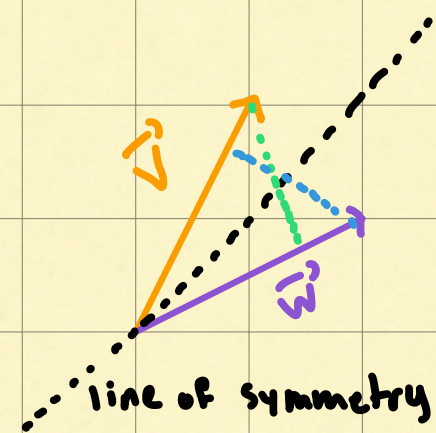


Proj $\vec{v}$ onto $\vec{w}$



$$|\vec{v}| \cdot |\text{Proj } \vec{w}_v| = \vec{v} \cdot \vec{w} = |\text{Proj } \vec{v}_w| \cdot |\vec{w}|$$

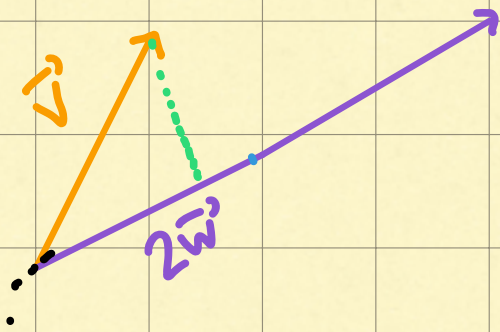
if $\vec{v}$ & $\vec{w}$ have the same length:



$$|\text{Proj } \vec{w}_v| \cdot |\vec{v}| = |\text{Proj } \vec{v}_w| \cdot |\vec{w}|$$

bc mirror image

if you scale one by constant... symmetry broken



$$\begin{aligned}\vec{v} \cdot 2\vec{w} &= |\text{Proj}_{\vec{w}} \vec{v}| \cdot |2\vec{w}| \\ &= 2(\vec{v} \cdot \vec{w})\end{aligned}$$

when you scale $\vec{w}$ by 2, doesn't change length of the projection,

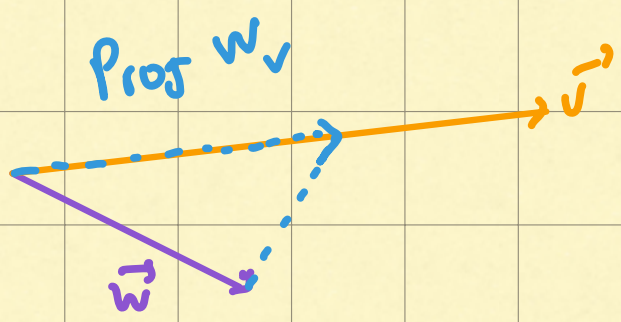
but doubles length of vector

you're projecting onto,

other way around, proj length changed, vector projecting onto unchanged

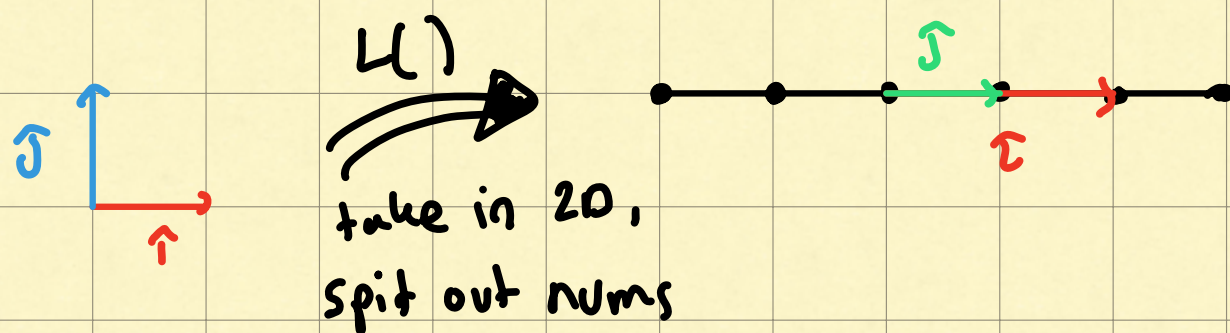
• order doesn't matter. bc scaling 1, just scale up.

• why are two processes connected?

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} d \\ e \\ f \end{bmatrix} = ad + be + cf \quad \stackrel{?}{\iff} \quad \text{Proj } w_v$$


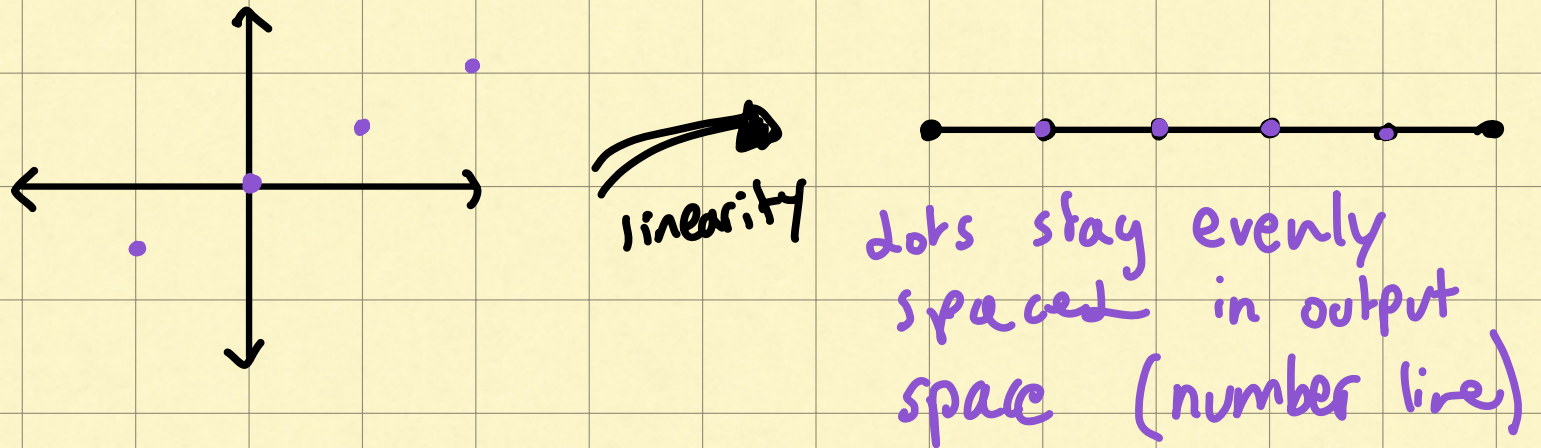
• duality.

• linear transformations from multiple dimensions to one dimension



$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow L(\vec{v}) \rightarrow [1.8]$$

- linear if line of evenly spaced dots in 2D, we still evenly spaced in 1D.

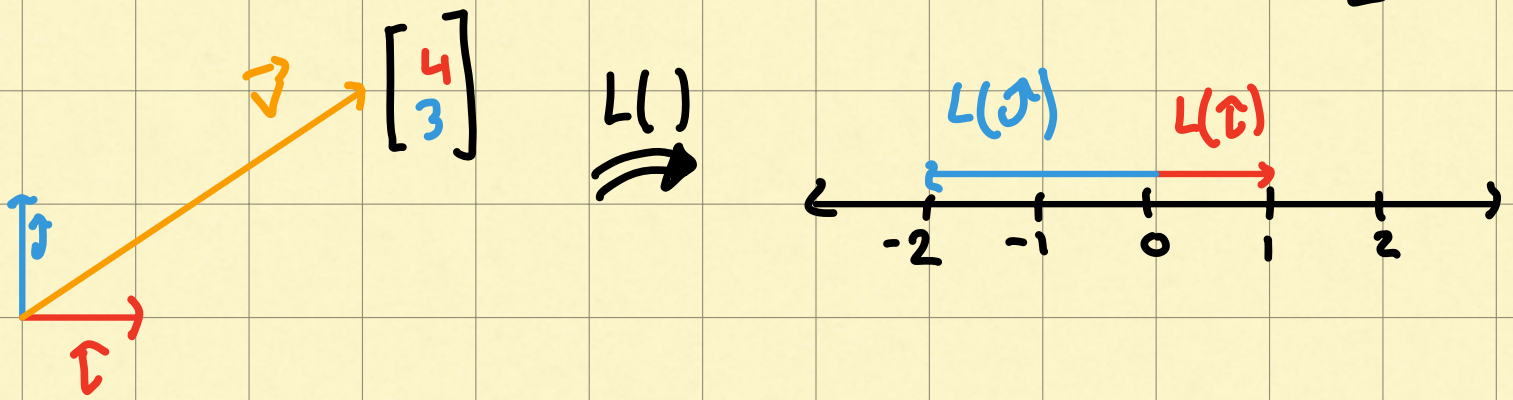


- linear transformations are determined by where $\hat{i}$ & $\hat{j}$ land. now they land on a $\#$

- transformation matrix $\begin{bmatrix} 2 & 1 \end{bmatrix}$
 $\hat{i}$ landed on 2 $\hat{j}$ landed on 1

- 1 row bc they land on one $\#$ each

example: $L()$ takes $\hat{i}$ to 1, $\hat{j}$ to -2: $\begin{bmatrix} 1 & -2 \end{bmatrix}$



vector w/ coordinates $\begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 4\hat{i} \\ 3\hat{j} \end{bmatrix}$.

$$\text{after } L() = 4 \cdot L(\hat{i}) + 3 \cdot L(\hat{j}) = -2$$

$$\begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{matrix} 4\hat{i} \\ 3\hat{j} \end{matrix} \xrightarrow[\text{L}()]{\text{linearity}} \begin{matrix} 4(L(\hat{i})) \\ 3(L(\hat{j})) \end{matrix} \begin{matrix} 4 \cdot \text{new } \hat{i} \\ 3 \cdot \text{new } \hat{j} \end{matrix}$$

$\vec{v} \qquad \qquad \vec{v}'$

• but this the same as matrix-vector multip

$$\begin{bmatrix} 1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 1 \cdot 4 + -2 \cdot 3 = -2$$

$\text{2D} \rightarrow \text{1D transform} \qquad \qquad \vec{v}$

so applying a linear transform from 2D to 1D

= matrix-vect multiplication

= same as taking dot product of 2 v's

• the $[\cdot \cdot]$ transformation matrix = vector
we tipped on it's side.

• numerically this is matrix vector mult.

• multiply 1×2 matrix by a 2D vector
feels just like taking dot product
of two vectors.

$$\begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

transformation vector

= dot product

- nice association b/w 1×2 matrices & 2D vectors

1×2 matrices $\longleftrightarrow$ 2D vectors

$$\begin{bmatrix} a & b \end{bmatrix} \longleftrightarrow \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} 2 & 7 \end{bmatrix} \longleftrightarrow \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

- going back & forth b/w matrices

& vectors means there is a

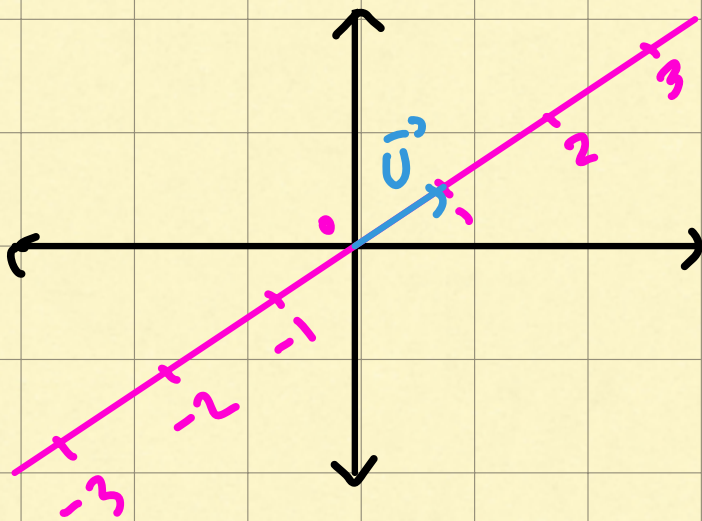
connection b/w linear transformations
that take vectors to numbers & vectors themselves

- geometrically: connection b/w lin tra.

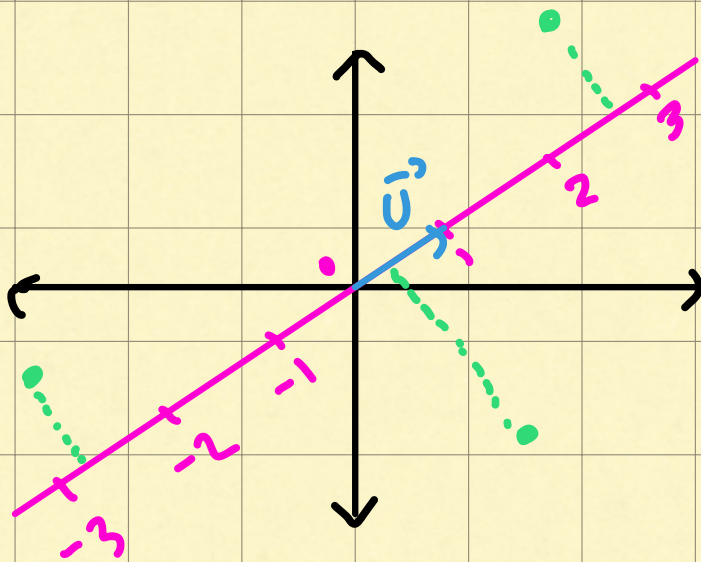
that take vectors to numbers & vectors themselves

- example of connection b/w
 - 2D → 1D lin transforms
 - vectors

- example: ~~x~~ line diagonal in 2D space



$\vec{u}^2$ = 2D unit vector, tip @ 1 on ~~x~~ line



Project 2D vectors onto ~~x~~ line

• = tip of vector

you've now defined a function that takes 2D vectors to numbers

• this function is also linear

• $[\cdot] \xrightarrow{4\vec{v}} [\cdot]$ • outputs are ~~xs~~,
not 2D vectors

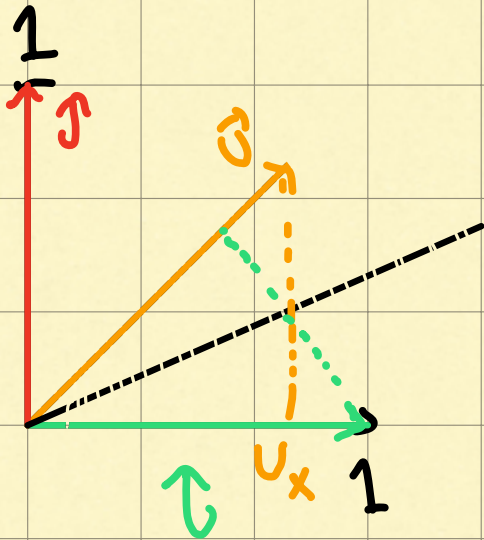
• the dots on ~~xx~~ line are numbers,
 $\vec{U}$ is a 2D vector overlapping
w/ embedded ~~xx~~ line.

• this projection is a linear
transformation from 2D vectors
to numbers. $[\cdot \cdot]$ projection
matrix

- can find a 1×2 matrix that describes this transformation.

where do $\hat{i}$ & $\hat{j}$ land?

bc unit vectors are columns of the matrix.



Proj $\vec{u}$ onto $\hat{i}$
Same as Proj $\hat{i}$ on $\vec{u}$

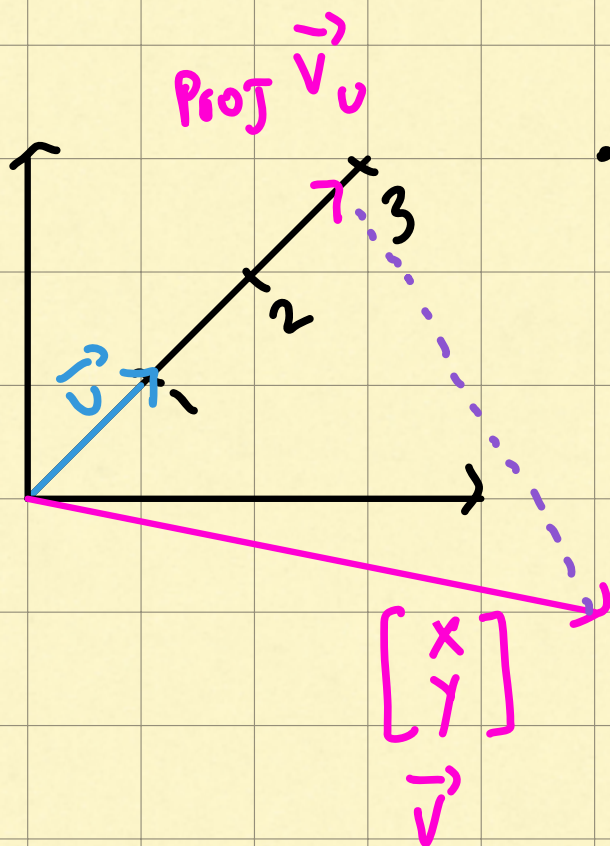
Q: what ~~x~~ does $\hat{i}$ land on when it gets projected?

A: same as what $\vec{u}$ lands on when it gets projected onto $\hat{i} = u_x!$

• Same reasoning. $\vec{J}$ lands on U_y .

• $\hat{i} \Delta \hat{j}$ land on $[U_x U_y]$
for $2D \rightarrow 1D$ linear transformation.
(Projection onto embedded ~~2~~ line).

• computing projection transformation
for arbitrary vectors in space:



• apply the Proj transform,

$$\begin{bmatrix} U_x & U_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Proj. trans.

$$= U_x X + U_y Y$$

computationally identical to
taking $\vec{v} \cdot \vec{u}$.

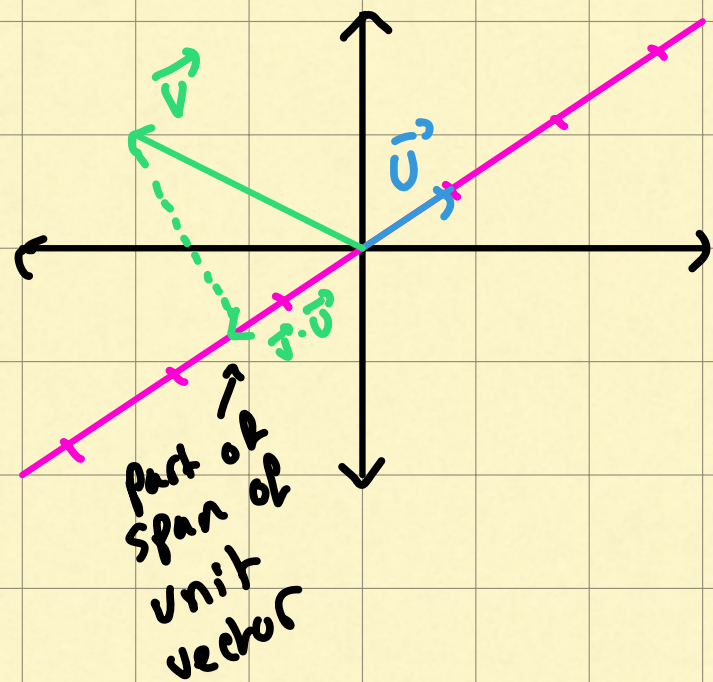
$\vec{u} \cdot \vec{v} \Leftrightarrow \text{Proj lin transform } (2D \rightarrow 1D)$

$$\begin{bmatrix} u_x & u_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = u_x x + u_y y$$

matrix-vector product

$$\begin{matrix} \updownarrow \\ \text{dot product} \end{matrix} \begin{bmatrix} u_x \\ u_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = u_x x + u_y y$$

.this is why taking dot product w/
Unit vector can be interpreted
as projecting a vector onto the
span of that unit vector & taking
the length



non unit vectors?

example: $\begin{bmatrix} 3u_x \\ 3u_y \end{bmatrix}$

matrix associated w/ vector: $\begin{bmatrix} 3u_x & 3u_y \end{bmatrix}$

takes $\hat{i}$ & $\hat{j}$ to $3u_x$ & $3u_y$.

• new matrix:

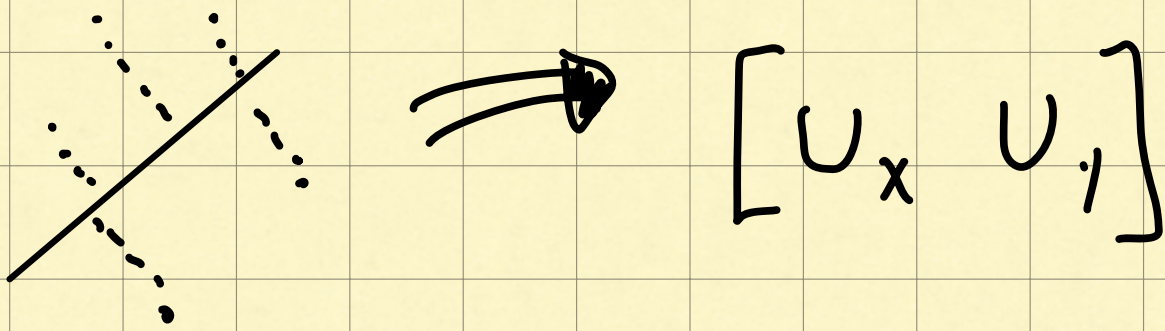
→ proj any matrix onto ~~x~~ line
→ multiply where it lands by 3

. dot product w/ non unit vector:

- project onto vector
- scaling up proj length by length of vector

• we had a linear transformation from 2D space to ~~a~~ line, not defined in terms of numerical vectors or dot products, but defined by projecting space onto a diagonal copy of the number line

• bc transformation is linear, it was necessarily described by some 1×2 matrix



- Since multiplying 1×2 matrix by 2D vector, is same as turning matrix on its side & taking dot product, this transformation was, related to some 2D vector

• anytime you have one of these linear transformations whose output space is the x line...

• there will be some unique vector $\vec{v}$, corresponding to that transformation

• applying transformation, is same as taking dot product w/ that vector.

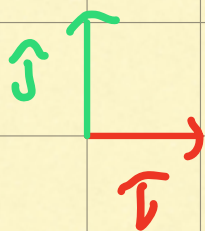
matrix-vector product

$$[u_x \ u_y] \begin{bmatrix} x \\ y \end{bmatrix}$$

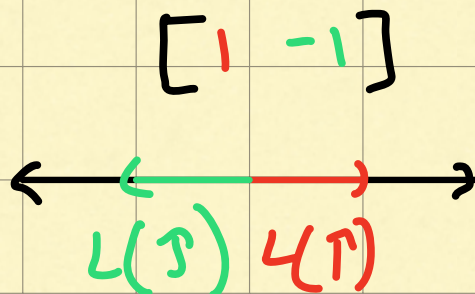
$\Downarrow$

Dot Product

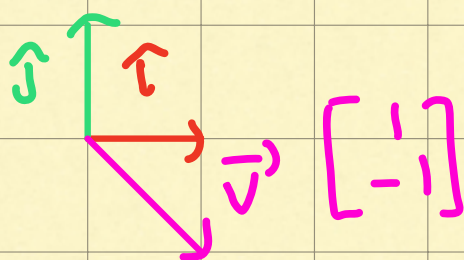
$$[u_x \ u_y] \cdot \begin{bmatrix} x \\ y \end{bmatrix} =$$



$L()$
 $\Rightarrow$



linear transform who's output
space is the number line



there will be some vector $\vec{v}$,
corresponding to that transformation

applying transformation $\begin{bmatrix} 1 & -1 \end{bmatrix}$
same as dot product: $\begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot [\dots]$

- any $2D \rightarrow 1D$ lin transform ... it's associated w/ some vector
- duality = correspondence b/w math things
- dual of a vector = lin transform it encodes
- dual of lin trans to $1D$, = vector in that space
- Summary = dot product = geometric tool for understanding projections

- & testing if vectors point in
Same direction

- joining 2 vectors together =
Translating 1 of them
into world of translations

- Sometimes easier to understand
vector not as arrow in space,
but as physical embodiment
of a linear transformation

• vector is conceptual shorthand
for certain transformations

• easier to think of arrows
in space, than moving space
to number line

• dot product tells you how much a vector is
working for/against the other, cosine similarities.